

# Forensic Log-Triage: Model Comparison

How well do language models detect malicious activity in Windows event logs?

FORCAST-L benchmark · generated 2026-06-18 · scoring contract `forecastl_scoring_20pt_v1` · all models scored against one frozen ground-truth snapshot

Each model reads raw Windows event-log records (EvtxECmd CSV) for a single forensic artifact and must decide **malicious vs. benign** and extract the supporting evidence. We measure three things that matter operationally: how many real attacks it catches (**recall**), how often it cries wolf on benign activity (**false-positive rate**), and how often it invents evidence that isn't in the log (**hallucination**). Lower is better for the last two.

## Results — same 120 questions, every model

A comparison is only meaningful when every model is asked the **same question on the same evidence**. Each model here saw the **identical prompt** ( `simple_malicious_prompt_v1` ) on the **same 120 artifacts** (60 attack / 60 benign) with identical ground truth and scoring — only the model differs. Best detection rate is highlighted; false-positive and hallucination rates are shown without a "winner" because the lowest value usually reflects a model that simply flags less.

| MODEL                                                                                        | ATTACK / BENIGN<br>SAMPLES | DETECTION<br>(RECALL)                       | FALSE-POSITIVE<br>RATE               | HALLUCINATION<br>RATE |
|----------------------------------------------------------------------------------------------|----------------------------|---------------------------------------------|--------------------------------------|-----------------------|
| <b>Claude Opus 4.8</b><br><small>Anthropic (frontier) · <code>claude-opus-4-8</code></small> | 60 / 60                    | <b>96.7%</b><br><small>95% CI 89–99</small> | 20.0%<br><small>95% CI 12–32</small> | 0.7%                  |
| <b>GPT-5.5</b><br><small>OpenAI (frontier) · <code>gpt-5.5-2026-04-23</code></small>         | 60 / 60                    | 91.7%<br><small>95% CI 82–96</small>        | 15.0%<br><small>95% CI 8–26</small>  | 0.6%                  |
| <b>Qwen3.6-27B</b><br><small>Local open-weight (27B) · <code>qwen3.6-27b</code></small>      | 60 / 60                    | 76.7%<br><small>95% CI 65–86</small>        | 11.9%<br><small>95% CI 6–22</small>  | 0.5%                  |

## What the benchmark is

- **328 real artifacts** — genuine Windows event logs, no synthetic data.
- Attacks span **11 MITRE ATT&CK tactics, 60 techniques**; benign set is real baseline/admin activity shaped as near-misses.
- Evidence scored across 6 fields (event IDs, processes, accounts, commands, network, registry) on a 20-point rubric.
- **Identical question for every model** — one frozen prompt template, parameterized only by the artifact; nothing is tailored per model.
- Deterministic: models run at **temperature 0**.

## Extraction quality

- **Claude Opus 4.8**: 95/118 files graded A (81%), mean score 18.0/20
- **GPT-5.5**: 85/119 files graded A (71%), mean score 17.4/20
- **Qwen3.6-27B**: 81/108 files graded A (75%), mean score 17.7/20

Grade A =  $\geq 17/20$ . High recall with low hallucination means the evidence a model reports can be trusted, not just its verdict.

## How to read this — and what it does *not* say

- **Single run per model.** Temperature 0 makes each run deterministic, but these are not yet averaged across repeated runs; treat differences inside the overlapping 95% confidence intervals as ties.
- **Coverage differs in the full-corpus table.** The local model was scored on the 120-file Standard slice, not all 328 — use the head-to-head table above for any apples-to-apples claim.
- **A "false positive" here is strict.** The benign set is deliberately adversarial (admin actions that look attack-shaped), so double-digit FP rates are expected and do not indicate a broken model.
- This is an independent capability comparison, not a vendor endorsement.

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Corpus provenance: attack artifacts from the public mdecrevoisier/EVTX-to-MITRE-Attack dataset (CC0); benign baseline from NextronSystems/evtx-baseline (Apache-2.0); both converted to CSV with EvtxECmd. Scoring contract forecastl\_scoring\_20pt\_v1 (frozen). Numbers are computed by replaying each model's saved responses through the production scoring pipeline against one common ground-truth snapshot; the replay is parity-verified against the original scored manifests. Recall/FP confidence intervals are 95% Wilson; errored-benign files are excluded from the FP denominator and an errored attack counts as a miss. Sample representativeness: on the full 328-file corpus, Claude Opus 4.8 95.4%, GPT-5.5 94.5% recall — consistent with the 120-file results above, indicating the shared set is representative for those models. The local model was evaluated on the 120-file set only.